

Blacklight Studio Works: Improving real- time analytics for mobile gaming with BigQuery

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[About Blacklight](#)

Blacklight Studio creates social games played by millions daily. With Google Cloud, they deliver a reliable experience and a thriving game ecosystem by analyzing real-time events with BigQuery.



About Blacklight Studio Works

Founded in 2012, Blacklight Studio Works brings people together with games. Its flagship game, Ludo SuperStar, is one of the most popular mobile board game apps in India with more than 100 million downloads since its launch in 2017. Blacklight attracts more than 2 million players daily with its library of popular games including CallBreak, Carrom, and Solitaire.

Industries: Gaming

Google Cloud results

- Zero downtime despite 300% increase in load after COVID-19 lockdown began
- Reduces game development time by more than 30%
- Keeps user data safe with Firebase Authentication, regardless of location
- Processes terabytes of data in seconds to better understand users and their interests with BigQuery
- Google Cloud enables a 100% work-from-home

Support
increased
downtime
with
downtime

Location: India

Engine (<https://cloud.google.com/appengine/>),



About Searce

Founded in 2004, Searce is a Google Cloud Partner that specializes in work transformation, infrastructure and app modernization, managed cloud services, and business-process automation leveraging Google Cloud services. Beyond consulting services, Searce provides hands-on engineering support to help more than 3,000 companies successfully migrate to Google Cloud. Searce has been recognized for its exceptional service as Google Cloud Partner of the Year for four consecutive years.

Google Cloud (<https://cloud.google.com/>)

BigQuery (<https://cloud.google.com/bigquery/>)

App Engine (<https://cloud.google.com/appengine/>)

Cloud Firestore
(<https://cloud.google.com/firestore/>)

Cloud Functions
(<https://cloud.google.com/functions/>)

Cloud SQL (<https://cloud.google.com/sql/>)

modernization, managed cloud services,
and business-process automation

policy during the pandemic

Firebase Realtime Database

(<https://firebase.google.com/products/realtime-database/>)

Firebase Authentication

(<https://firebase.google.com/products/auth>)

Looker Studio

(<https://marketingplatform.google.com/about/data-studio/>)

Google Marketing Platform

(<https://marketingplatform.google.com/about>)

Millions of people play games on mobile devices every day. According to Statista, the casual game genre, which includes games like Angry Birds, accounts for **78% of all mobile gaming** (<https://www.appannie.com/en/go/gaming-2021/>)

in 2020. Shruti Sarraf, co-founder of the gaming company **Blacklight Studio Works**

(<https://blacklightsw.com/>) points out that mobile board games such as its flagship games, Ludo Superstar and Carrom Superstar, are easy to play and connect people with friends and family. As a result, these games attract fans, from Brazil to Saudi Arabia, who play and interact socially while gaming online.

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75% of the studio's game downloads stem from India, where increasing smartphone adoption means more people are playing mobile games. To meet the growing appetite for mobile games, the studio leverages Google Cloud (<https://cloud.google.com/>) to gain valuable user insights and build better games.

"Analytics offer insights that are critical to game developers. By looking at data, we learn more about our players," says Sarraf. "BigQuery ([/bigquery](https://cloud.google.com/bigquery/)) is a game changer because we can track all our data to understand what people want and make data-informed decisions to improve our games."

Blacklight started its Google Cloud journey in 2017 with Firebase (<https://firebase.google.com/>) for game

development when the company pivoted their focus towards building real-time multiplayer games. The studio now uses Firestore Realtime Database (<https://firebase.google.com/docs/database>) along with Cloud Functions (/functions) to build the core multiplayer backend, Cloud SQL (/sql) for storing game and player data, and BigQuery for real-time gaming analytics, among other Google Cloud products.

It also works with Google Cloud Premier Partner Searce (<https://www.searce.com/>), which provides regular training and updates to keep Blacklight up to speed on the latest Google Cloud products and services.

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—*Shruti Sarraf, co-
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Works*

Getting smarter analytics to improve player engagement and in-app revenue

Creating a viable game economy where players can collect and win game currency contributes to the success of free-to-play games. Blacklight depends on in-app purchases and advertising revenue to make money and continue building popular games. If the game is too complex too early, players may leave instead of buying in-app purchases to level up. Likewise, if you give players too many free coins, they won't spend money inside the game.

To answer the pressing question, "What's the best reward system to keep users coming back?" the studio must process and analyze vast amounts of data, from app-performance data to user behavior. So as the number of daily players increased from thousands to millions of users, Blacklight turned to BigQuery, built for scale and processing terabytes of data in seconds.

"BigQuery helps us overcome the limitations of our legacy SQL server," says Sarraf. "With the seamless

scalability of BigQuery, we can query billions of rows to get the right data in just a few seconds."

The studio uses BigQuery to link half a gigabyte of event data from Firebase that tracks user interaction, with half a gigabyte of non-real-time streaming data such as historical archives in Cloud SQL to compare performance over time. Based on the insights, the studio can automatically trigger events based on user behavior to improve the game economy. For instance, they offer a specific reward level to a player based on the number of coins they have.

Blacklight also uses Looker Studio

(<https://marketingplatform.google.com/about/data-studio/>)

to create reports and charts to visualize BigQuery data.



Accelerating app development with automated tools

Compared to console and PC games, mobile games have a shorter development time and require regular

updates to keep players interested. The team at Blacklight keeps busy with one major release every two months and updates twice a month to improve stability and performance. Now it is able to further reduce its development time by more than 30%.

The studio uses Firebase to do more with its lean team, since they do not have to worry about manual coding. For example, when the studio plans a release or update, it uses [Firebase Remote Config](https://firebase.google.com/products/remote-config) (<https://firebase.google.com/products/remote-config>), a developer tool to push the application to a small percent of users and set performance perimeters to see what works well.

Blacklight further minimizes risks of its app performance by doing A/B testing, whereby it compares two different scenarios of a controlled setting, through Firebase, to find out which app version works better before releasing the live app.

To keep user data safe, the studio uses [Firebase Authentication](https://firebase.google.com/docs/auth) (<https://firebase.google.com/docs/auth>) to identify users on the app and [Cloud Functions](#) ([/functions](#)) for secure access to the [Firebase Realtime Database](#) (<https://firebase.google.com/docs/database>) based on the user's identity.

Supporting remote work during COVID-19

In 2020, Blacklight experienced a 300% increase in downloads and daily active users, mainly due to

more people staying at home because of COVID-19 restrictions.

"Internally, we switched to a 100% work-from-home policy to stay safe and relied on Google Cloud's managed services capabilities to deliver uninterrupted gaming," says Sarraf. "Despite the change in working environment and increase in downloads, we could still remain focused on developing new games and didn't have to worry about managing capacity for traffic spikes. This is because Google Cloud products like App Engine and Cloud Functions automatically spin up new instances if the workload exceeds capacity."

Following training from Searce, Blacklight set up alerting policies to notify the tech team about incidents that require attention, such as low memory states and CPU.

Sarraf shares that the training, tutorials, and use cases of Google Cloud make onboarding new hires a breeze. There are numerous guides to help engineers who may be new to Google Cloud understand how to use and connect to Google Cloud services.

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The modular services interact with each other seamlessly and securely with minimum coding effort. I believe the wide range of Google Cloud products can help us expand our reach in the mobile games market with scalable cloud services and smarter game analytics."

—*Shruti Sarraf, co-founder, Blacklight Studio Works*

Scheduling tasks efficiently to improve the game experience

Moving forward, Blacklight is looking at [Cloud Tasks \(/tasks\)](#) for scheduling specific tasks such as a daily event that runs periodically. For example, Ludo

Superstar offers time modes such as Blitz and Gold, where players receive bigger prizes when they log in at a specific time.

"Google Cloud is designed with users in mind.

Bringing our frontend and backend services together makes things very secure and simple. The modular services interact with each other seamlessly and securely with minimum coding effort," says Sarraf. "I believe the wide range of Google Cloud products can help us expand our reach in the mobile games market with scalable cloud services and smarter game analytics."

